

IPv4 Header and Fragmentation: Architectural Analysis and Technical Specifications

Executive Summary

The Internet Protocol Version 4 (IPv4) operates at the Network Layer and is responsible for logical addressing, routing, and fragmentation of datagrams.

Key architectural insights:

- **Header Size:** Variable from 20 bytes to 60 bytes.
 - **Scaling Mechanism:** Header Length (HLEN) uses a scaling factor of 4.
 - **Service Differentiation:** Type of Service field enables prioritization.
 - **Fragmentation Control:** Managed through Identification, Flags (DF, MF), and Fragment Offset.
 - **Efficiency Impact:** Fragmentation reduces transmission efficiency due to header replication.
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1. Network Layer Context and Encapsulation

In the TCP/IP model:

- Application Layer → Message
- Transport Layer → Segment (TCP) / User Data (UDP)
- Network Layer → Datagram (IPv4 packet)

At the Network Layer, the IPv4 header is added to create a datagram. This header contains routing and control information required for delivery across interconnected networks.

2. IPv4 Header Structure and Size Management

The IPv4 header consists of:

- A fixed 20-byte portion
- An optional portion (0–40 bytes)

2.1 Fixed and Variable Components

Fixed Header

- 5 rows
- Each row = 32 bits (4 bytes)
- Total fixed size = 20 bytes

Options Field

- Size range: 0 to 40 bytes
- Used for special routing or control requirements

Size Limits

- Minimum header = 20 bytes
- Maximum header = 60 bytes

2.2 Header Length (HLEN) and Scaling Factor

HLEN is a 4-bit field.

Since 4 bits can represent values from 0 to 15:

Maximum representable value = 15

To represent up to 60 bytes:

Scaling factor = 4

HLEN value = Header Size / 4

Header Size (Bytes)	HLEN Value
20	5
32	8
40	10
60	15

Padding Rule

If header size is not a multiple of 4:

- Padding bytes are added.
 - Padding ensures 32-bit alignment.
 - Receiver discards padding.
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3. Core IPv4 Header Fields

3.1 Version (4 bits)

Indicates IP version:

- IPv4 → 0100
 - IPv6 → 0110
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3.2 Type of Service (ToS) – 8 bits

Used for Quality of Service control.

Structure:

- First 3 bits → Priority (0–7)
- Next 4 bits → Service Type
 - D → Minimize delay
 - T → Maximize throughput
 - R → Maximize reliability
 - C → Minimize cost
- Last bit → Reserved

Rule:

Only one of D, T, R, C should be set at a time.

Routers may drop lower-priority packets during congestion.

3.3 Total Length (16 bits)

Represents:

Header + Data

Maximum value:

$$2^{16} - 1 = 65,535 \text{ bytes}$$

Maximum data payload (minimum header):

$$65,535 - 20 = 65,515 \text{ bytes}$$

4. Fragmentation Mechanics

Fragmentation occurs when:

Packet Size > MTU of outgoing link

4.1 Identification (16 bits)

- Unique value assigned to each original datagram.
 - All fragments carry the same Identification number.
 - Used for reassembly at destination.
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4.2 Flags (3 bits)

Bit	Purpose
Bit 0	Reserved
Bit 1 (DF)	Don't Fragment
Bit 2 (MF)	More Fragments

DF = 1

- Router cannot fragment.
- If packet exceeds MTU → discarded.
- ICMP error sent to source.

MF = 1

- More fragments follow.

MF = 0

- Last fragment (or only fragment).
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4.3 Fragment Offset (13 bits)

Indicates:

Position of fragment data relative to original datagram.

It represents:

Number of data bytes preceding this fragment.

Used by receiver to:

- Reassemble fragments
 - Handle out-of-order arrival
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5. Fragmentation and Efficiency Impact

Fragmentation increases overhead because:

- Each fragment carries a full IPv4 header.
- Header must be replicated.

Example Scenario

Original data = 300 bytes

Header = 20 bytes

MTU = 120 bytes

Each fragment:

120 bytes total → 20 header + 100 data

Fragments required:

$300 / 100 = 3$ fragments

Total header overhead:

$3 \times 20 = 60$ bytes

If unfragmented:

Header overhead = 20 bytes

Efficiency Comparison

Unfragmented

Efficiency = Data / Total
= $300 / 320$
 $\approx 93.75\%$

Fragmented

Total transmitted = 300 data + 60 header = 360

Efficiency = $300 / 360$
 $\approx 83.33\%$

6. Architectural Insight

IPv4 is connectionless:

- Each fragment is routed independently.
- Fragments may take different paths.
- Each fragment must carry full routing information.

Thus, header replication is mandatory for independent delivery.

Final Technical Perspective

IPv4 header design balances:

- Addressing flexibility
- Routing control
- Fragmentation capability
- Backward compatibility

However, fragmentation introduces inefficiency due to:

- Header duplication
- Increased processing load
- Higher probability of retransmission (if one fragment is lost)

Modern networks therefore attempt to avoid fragmentation through:

- Path MTU discovery
- Proper packet size configuration

Understanding these structural and mathematical relationships is essential for competitive exams and real-world network optimization.

